SDM COLLEGE OF ENGINEERING AND TECHNOLOGY, Dharwad-580002

Department of Computer Science and Engineering

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IoT [22UCSE521] – CTA

5th Semester B.E (CSE)

Title:

“Driver Drowsiness Detection System Using Raspberry Pi”

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1. Introduction:

Road accidents caused by driver fatigue account for a significant percentage of traffic fatalities worldwide. Long driving hours, inadequate sleep, and monotony of highways often result in drowsiness. A drowsy driver experiences reduced reaction time, blurred vision, involuntary eye closure, and micro-sleeps, which dramatically increase the risk of severe crashes.

To address this issue, intelligent driver monitoring and fatigue detection systems have gained importance in modern automotive safety. This project presents a **real-time Driver Drowsiness Detection System** developed using **Raspberry Pi 4**, a **USB camera**, and **computer vision techniques**.

The system continuously monitors the driver's eyes using **live video stream analysis**. Using **dlib's 68 facial landmark model**, the Eyed Aspect Ratio (EAR) is calculated to determine whether the driver's eyes are open, partially closed, or fully closed.

The key features include:

- **Real-time detection of drowsiness using EAR calculation**
- **Non-intrusive monitoring using external USB camera**
- **Immediate visual alert using WS2813 RGB LED stick**
- **Audible alarm using buzzer module**
- **Voice alert using eSpeak to wake the driver**

This system is low-cost, easy to implement, and can be integrated into vehicles for enhanced road safety. It serves as a practical demonstration of IoT, embedded systems, and AI-based computer vision.

2. Usage and Purpose of the Project:

The primary purpose of the **Driver Drowsiness Detection System** is to enhance road safety by continuously monitoring the behavioural patterns of a driver and identifying early signs of fatigue or micro-sleep. Drowsiness is one of the major contributors to road accidents, especially during long-distance highway driving. This system aims to address this issue with a low-cost, real-time, and non-intrusive solution.

I. Purpose of the Project

1. **To reduce road accidents caused by driver fatigue**
   By detecting early signs of drowsiness, the system alerts the driver before they lose control.
2. **To provide real-time monitoring using computer vision**
   Eye Aspect Ratio (EAR) based analysis ensures accurate detection of drowsiness without physical contact sensors.
3. **To offer an affordable safety enhancement**
   The system uses easily available components such as Raspberry Pi, USB camera, and buzzer, making it cost-effective compared to commercial driver monitoring systems.
4. **To demonstrate IoT and embedded system integration**
   It showcases the integration of hardware (LEDs, buzzer, camera) with AI-based software (OpenCV, dlib, landmark detection).
5. **To promote safe driving practices**
   The system acts as a reminder for drivers to rest at intervals, reducing stress and fatigue.

II. Usage of the Project

This project can be used in multiple real-world scenarios:

1. **Personal Vehicles**
   Acts as a driver safety assistant, especially during night drives or long journeys.
2. **Commercial Transport**
   Useful for truck drivers, bus drivers, cab drivers, or delivery personnel who work extended hours.
3. **Fleet Management Systems**
   Companies can integrate this system into fleet vehicles to monitor driver alertness and reduce accidents.
4. **Automotive Research and Training Institutes**
   Demonstrates the importance of driver alertness and showcases practical implementation of AI in vehicles.
5. **College and University Projects**
   A strong educational example of combining computer vision, IoT hardware, and safety-focused applications.
6. **Smart Vehicle Systems / ADAS Modules**
   Can be incorporated into Advanced Driver Assistance Systems for enhanced safety.
7. **Rental Vehicle Monitoring**
   Ensures drivers of rented self-drive cars remain alert.

III. Benefits

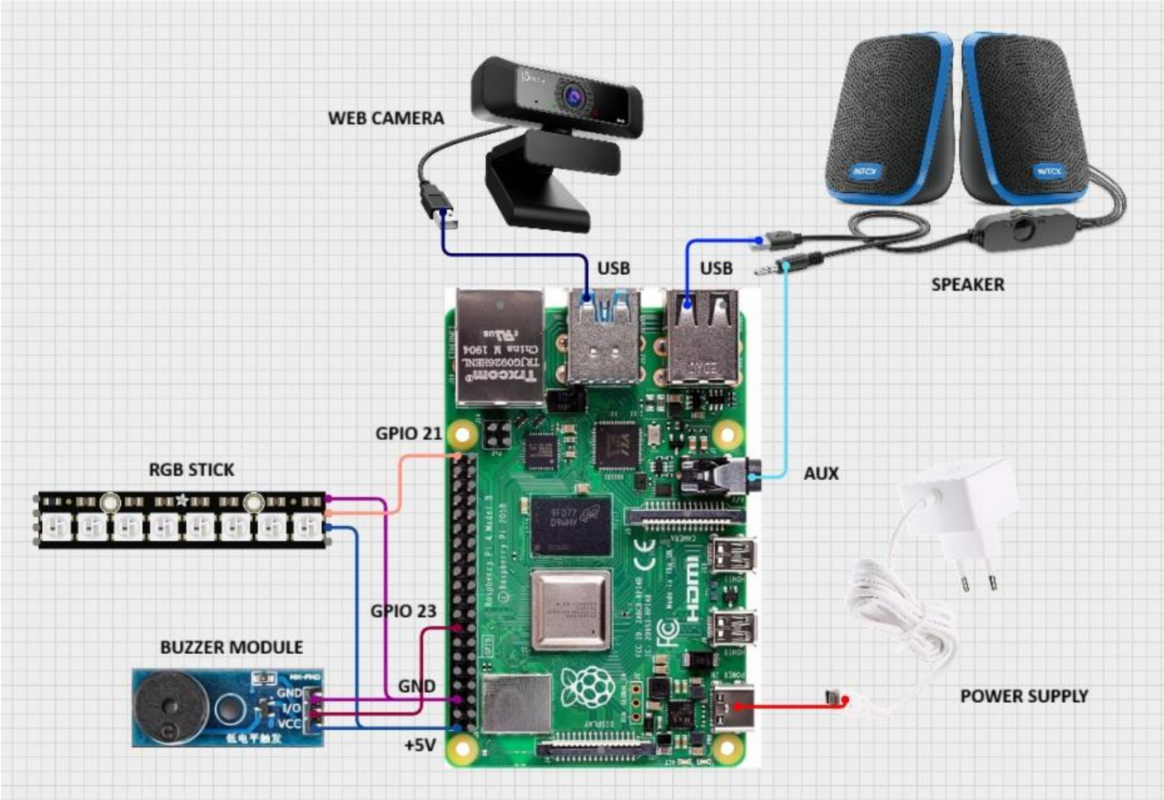
- Simple and affordable setup
- Real-time detection with high accuracy
- Non-intrusive (no wearable devices required)
- Audio + visual alerts for maximum effectiveness
- Can run continuously without overheating or system lag
- Highly customizable for different vehicle environments

3. Components Required:

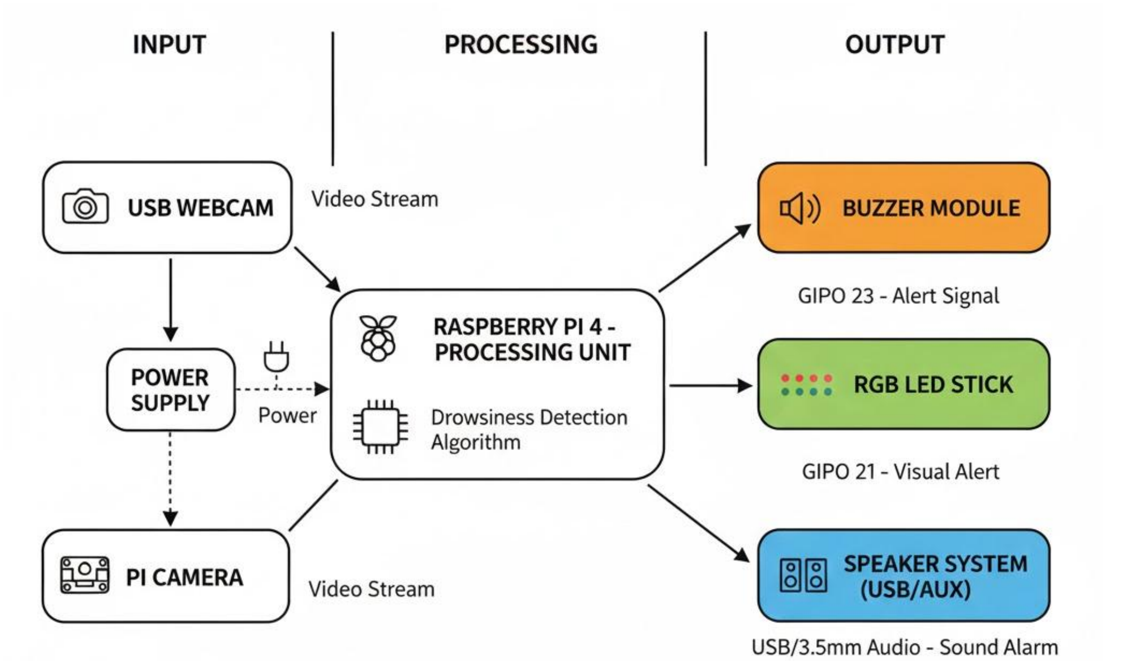
SL.NO	Component Name	Specification / Description
1.	Raspberry Pi 4 Model B	4GB/8GB RAM, main controller
2.	USB Webcam	External HD USB camera for real-time video
3.	WS2813 RGB LED Stick (10 LEDs)	NeoPixel-compatible addressable LED strip for status indication
4.	Active Buzzer Module	5V buzzer with VCC, GND, Signal pins
5.	MicroSD Card (16/32 GB)	Contains Raspberry Pi OS & project files
6.	5V 3A Power Adapter	Official Raspberry Pi USB-C power supply
7.	HDMI Monitor	Displays system output
8.	USB Keyboard & Mouse	For setup and configuration
9.	Jumper Wires (F-F / F-M)	For GPIO connections

4. Working Of Circuit:

➢ Circuit Diagram:



➢ Block Diagram:



The system integrates **hardware components** with **computer vision algorithms** to detect drowsiness.

I. Frame Acquisition

The external USB webcam captures real-time video frames.
OpenCV reads the frames and converts them to grayscale for fast processing.

II. Face Detection

Using **dlib’s HOG + Linear SVM-based frontal face detector**, the driver's face is localized inside each captured frame. This step identifies:

- Face boundary
- Orientation
- Proximity of the driver

III. Eye Landmark Detection

dlib’s **shape predictor model** (68_face_landmarks.dat) identifies 68 key facial landmarks.
Specific eye landmarks (points 36–41 for left eye, 42–47 for right eye) are extracted.

This enables precise localization of eye contours.

IV. Eye Aspect Ratio (EAR) Computation

EAR is calculated using vertical and horizontal eye landmark distances:

- **EAR high** → Eyes open
- **EAR medium** → Partially closed
- **EAR low** → Eyes fully closed

The EAR value is extremely stable and robust against:

- Head movements
- Blinking
- Lighting variations

V. Drowsiness Classification Logic

The system uses threshold-based classification:

EAR Value	Condition
> 0.25	Eyes Open (Active)
0.21–0.25	Partially Closed (Drowsy)
< 0.21 for continuous frames	Sleeping

Continuous low EAR for multiple frames indicates **micro-sleep or drowsiness**.

VI. Alert Mechanism

Based on the driver's state:

1. Eyes Open (Active State)

- GREEN light on WS2813 LED
- Buzzer OFF
- No voice alert

2. No Face Detected

- BLUE LED (indicating driver missing or not in view)
- Buzzer OFF
- System continues scanning

3. Drowsy or Sleeping State

- RED LED on WS2813
- Buzzer turns ON
- Voice alert plays: **"Wake up Driver!"**
- Alerts continue until EAR returns to normal

GPIO Connections:

- **Buzzer Signal → GPIO 23**
- **WS2813 DIN → GPIO 18**
- **Common GND → Raspberry Pi Ground Pin**
- **5V Power → Pi's 5V pin**

This combination ensures real-time reliability and quick response to driver fatigue.

5. Results:

The developed system was successfully able to:

- Detect the driver's face using a USB webcam
- Identify facial landmarks and determine eye openness
- Calculate EAR values accurately
- Trigger alerts during drowsy or prolonged eye-closure states
- Provide clear visual indication through RGB LED stick
- Produce sound alarms with buzzer
- Announce voice alert “Wake up Driver” using eSpeak
- Display real-time status on screen (Awake, Drowsy, No Face)

The system performed efficiently in real-time with minimal delay, making it suitable for vehicle safety applications.

6. Future Scope:

The Driver Drowsiness Detection System can be further improved in several ways:

1. **Night Vision Support:**
   Use of infrared or NoIR cameras to improve detection in low-light or night driving conditions.
2. **Mobile App Connectivity:**
   Integration with a smartphone app to send real-time alerts or notifications to fleet managers or family members.
3. **Vehicle Control Integration:**
   System can be connected to car controls to automatically slow down the vehicle or activate safety features if the driver is unresponsive.

4. **Deep Learning Enhancements:**
   Implementing advanced machine learning models for more accurate detection and classification of drowsiness.
5. **Cloud Data Logging:**
   Store driving and fatigue patterns in the cloud for long-term analysis and safety improvement.

These enhancements can make the system more reliable, intelligent, and suitable for commercial automotive applications.

7. Bill of Components:

Sl. No	Component	Quantity	Unit Price	Total Price
1	Raspberry Pi 4 Model B	1	₹5,000	₹5,000
2	USB Webcam	1	₹400	₹400
3	WS2813 RGB LED Stick (10 LEDs)	1	₹250	₹250
4	Active Buzzer Module	1	₹30	₹30
5	MicroSD Card (32GB)	1	₹300	₹300
6	Raspberry Pi 5V 3A Adapter	1	₹700	₹700
7	Jumper Wires (Set)	1	₹50	₹50

Total Estimated Cost: ₹6,880

8. References:

- Dlib Library Documentation – http://dlib.net
- OpenCV Official Documentation – https://docs.opencv.org
- Raspberry Pi GPIO Guide – https://www.raspberrypi.org/documentation/
- T. Soukupová and J. Cech, "Real-Time Eye Blink Detection using Facial Landmarks"
- Adafruit NeoPixel Guide – https://learn.adafruit.com
- Python RPi.GPIO Documentation